THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

CLASS VIII — MATHEMATICS · MID-TERM PRACTICE PAPER

Session 2026-27 | CBSE / RBSE Pattern

Class: 8 | Subject: Mathematics | Set 04 | Time: 2 Hours 30 Minutes | Max. Marks: 60

General Instructions:

1. This question paper contains 16 questions. All questions are compulsory.
2. Question paper is divided into FIVE sections – Section 'A', 'B', 'C', 'D' and 'E'.
3. In Section 'A' question number 1 has 12 multiple choice sub-parts carrying 1 mark each.
4. In Section 'B' question numbers 2 to 7 are very short answer type questions carrying 2 marks each.
5. In Section 'C' question numbers 8 to 10 are short answer (SA) type questions carrying 3 marks each.
6. In Section 'D' question numbers 11 to 13 are long answer (LA) type questions carrying 5 marks each.
7. In Section 'E' question numbers 14 to 16 are case-based questions whose sub-parts are of 1, 1 and 2 marks respectively.
8. There is no overall choice. However, internal choice has been provided in 1 question of Section B, 1 question of Section C, 2 questions of Section D, and in the 2-mark sub-part of every case-based question in Section E. Attempt only one of the alternatives in all such questions.
9. Draw neat figures wherever required. Use of calculators is NOT allowed. Write the serial number of the question before attempting it.

SECTION A — Multiple Choice Questions (12 × 1 = 12) — 1 questions · 12 marks

1. Question 1 consists of multiple-choice questions (i–xii) of 1 mark each. Choose the correct option.

(i) The square root of 0.0081 is: **1**

(a) 0.09 (b) 0.9 (c) 0.009 (d) 0.081

(ii) The cube of −4 is: **1**

(a) 64 (b) −64 (c) −16 (d) 16

(iii) The number of non-square numbers between 6^2 and 7^2 is: **1**

(a) 6 (b) 13 (c) 24 (d) 12

(iv) The value of $2^{-3} \times 2^5$ is: **1**

(a) 2 (b) 8 (c) 1/4 (d) 4

(v) The Roman numeral XLIV stands for: **1**

(a) 46 (b) 64 (c) 44 (d) 36

(vi) $(101)_2$ in the decimal system is: **1**

(a) 3 (b) 5 (c) 6 (d) 7

(vii) A quadrilateral with all sides equal and one angle 90° is a: **1**

(a) rhombus that is not a square (b) square
(c) kite (d) trapezium

(viii) The diagonals of every parallelogram: **1**

(a) are equal (b) are perpendicular
(c) bisect each other (d) are parallel

(ix) $(a + b)^2 - (a - b)^2$ equals: **1**

(a) 2ab (b) $2a^2 + 2b^2$ (c) 0 (d) 4ab

(x) 5 kg : 500 g in simplest form is: **1**

(a) 1 : 10 (b) 5 : 500 (c) 10 : 1 (d) 1 : 100

(xi) $x^2 - 49$ can be factorised as: **1**

(a) $(x - 7)^2$ (b) $(x + 7)^2$

(c) $(x - 7)(x + 7)$ (d) $(x - 49)(x + 1)$

(xii) If 3 : 5 :: x : 20, then x equals: **1**

(a) 15 (b) 8 (c) 10 (d) 12

SECTION B — Very Short Answer (6 × 2 = 12) — 6 questions · 12 marks

Questions 2 to 7 carry 2 marks each.

2. Find the square root of 2025 by prime factorisation. **2**

3. Is 392 a perfect cube? If not, find the smallest number by which it must be multiplied to get a perfect cube. **2**

4. Write 53,000,000 and 0.000345 in standard form. **2**

5. Write 2024 in Roman numerals. **2**

OR

Convert $(234)_5$ into a decimal number.

6. In rhombus ABCD, $\angle A = 70°$. Find $\angle B$ and $\angle C$. **2**

7. Expand $(x + 4)^2$ and $(3x + 2)(3x - 2)$. **2**

SECTION C — Short Answer (3 × 3 = 9) — 3 questions · 9 marks

Questions 8 to 10 carry 3 marks each.

8. Find the square root of 1.69 and the cube root of 0.512. **3**

9. The angles of a quadrilateral are in the ratio 2 : 3 : 3 : 4. Find each angle. **3**

OR

In parallelogram PQRS, $\angle P : \angle Q = 2 : 3$. Find all four angles.

10. Using identities, evaluate: (i) $(1.02)^2$ (ii) 48×52 (iii) 302^2 **3**

SECTION D — Long Answer (3 × 5 = 15) — 3 questions · 15 marks

Questions 11 to 13 carry 5 marks each.

11. (i) Find the smallest number by which 288 must be multiplied so that the product is a perfect square. Find the square root of the product.
(ii) Find the square root of 5476 by prime factorisation. **5**

OR

(i) Find the smallest number by which 1944 must be divided so that the quotient is a perfect cube. Find the cube root of the quotient.
(ii) The volume of a cube is 0.064 m^3. Find the length of its edge.

12. Simplify: (i) $5^{-3} \times 5^5$ (ii) $(2/3)^{-3}$ (iii) $7^6 \div 7^8$ (iv) $(x^{-2}y^3)^2$ (v) $3^0 + 4^{-1}$ **5**

OR

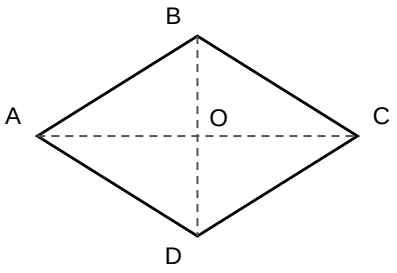
(i) Write 1994 in Roman numerals. (ii) Convert 45 to base 2. (iii) Convert $(212)_3$ to decimal. (iv) Add $(110)_2$ and $(111)_2$. (v) Convert 58 to base 8.

13. (i) Divide ₹ 2,400 among A, B and C in the ratio 3 : 5 : 4. **5**
(ii) 12 workers finish a job in 15 days. How many days will 20 workers take?
(iii) Find x if 4 : 9 :: 16 : x.
(iv) A train travels 240 km in 4 hours. How long will it take for 420 km at the same speed?
(v) A 40-litre mixture has milk and water in the ratio 7 : 3. Find the quantity of water.

SECTION E — Case-Based Questions (3 × 4 = 12) — 3 questions · 12 marks

Questions 14 to 16 carry 4 marks each (1 + 1 + 2).

14. A rhombus-shaped garden ABCD has diagonals AC = 16 m and BD = 12 m meeting at O. Each side of the garden is 10 m. ∠ABC = 100°.



(i) What is the angle between the diagonals AC and BD? 1

(ii) Find the length of AO. 1

(iii) Find the cost of fencing the garden at ₹ 50 per metre. 2

OR

(iii) Find ∠BAD and ∠OAB.

15. In a laboratory, a bacterial culture doubles every hour. Starting with 1 bacterium, the number after n hours is 2^n.

(i) How many bacteria are there after 5 hours? 1

(ii) How many bacteria are there after 10 hours? 1

(iii) After how many hours will the count reach 4096? 2

OR

(iii) How many times more bacteria are there at 12 hours than at 9 hours?

16. A fruit punch is made by mixing orange juice, pineapple juice and soda in the ratio 4 : 3 : 1. A party needs 4 litres of punch.

(i) What is the total number of parts in the mixture? 1

(ii) How much orange juice is needed for 4 litres of punch? 1

(iii) How much soda is needed to make 10 litres of the same punch? 2

OR

(iii) Only 4.5 litres of pineapple juice is available. What is the maximum amount of punch that can be made keeping the same ratio?